Department of Information Systems and Technologies

CTIS221 – Object Oriented Programming

FALL 2026 - 2027

Lab Guide 5 - Week 5 - 1

OBJECTIVES : Arrays
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Q1. Write a Java program that finds the average of 5 numbers in the array that is filled by the numbers that user entered. The program should also calculate the sum and the average of these numbers and the difference of each number to the average of all numbers, then displays the output messages and a table like below on the screen.

Example Run:

```
Enter 5 integer numbers:
23
14
56
23
12

The array content:
23 14 56 23 12

Sum of the numbers: 128. Average of the numbers: 25.60

index  number  diff. to avg
-----  ------  ------------
    0      23         -2.60
    1      14        -11.60
    2      56         30.40
    3      23         -2.60
    4      12        -13.60
```

Q2. Write a Java program that displays the given menu below and calls the appropriate method for the selected menu item. The array size is 10.

Write the following static methods;

- **menu:** display the menu on the screen, then read, validate and return the user's choice.

```
          MENU
   1. Initialize array
   2. Input array elements
   3. Output array elements
   4. Search a value
   5. Exit program
```

- **initialize:** sets all integer elements of the array to zero.
- **fillArray:** reads the integer values from the user until zero is entered or the array size is reached. The method must return the number of actual elements inputted as well as the array itself.
- **displayArray:** receives an integer array and the number of actual elements as parameters. Display all array elements on the screen.
- **findPos:** searches for an integer value in the array and returns that position (index) of the first occurrence of the value and returns **-1** if the value is not present.

Example Run:

```
       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 1
All elements are initialized by zero
```

```
       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 3
0 0 0 0 0 0 0 0 0 0

       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 9
Enter your choice: 8
Enter your choice: 2

Enter the elements of the array:
10
20
30
40
50
0
The size of the array: 5


       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 3
10 20 30 40 50


       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 4
Enter the value that will be searched: 20
Position of the value is: 1

       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 4
Enter the value that will be searched: 99
The array does NOT contain the searched number!

       MENU
1. Initialize array
2. Input array elements
3. Output array elements
4. Search a value
5. Exit program

Enter your choice: 5
```

Additional Questions

AQ. Write a Java program that displays the given menu below and calls the appropriate methods for the selected menu items. First ask user to fill the array with the integer numbers and then ask users' choice from the menu. The integer array must have maximum 100 elements. Depending on the choice, perform the operation mentioned in the menu item selected, write each menu item as a separate method.

Write the following static method;

- **menu:** displays the menu on the screen, then reads, validates and returns the user's choice.

```
DISPLAY
1.  All numbers
2.  Even numbers
3.  Subscripts (indices) of odd numbers
4.  The numbers with even subscripts
5.  Minimum number
6.  Subscript of maximum number
7.  Exit
```

Example Run:

```
Enter the elements of the array: 25 14 25 54 45
69 74 58 78 52 59 14 145 25 36 14 475 111 25 44
0

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 9

Enter your choice: -5

Enter your choice: 1

All numbers
************
25 14 25 54 45 69 74 58 78 52 59 14 145 25 36
14 475 111 25 44

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 2

Even numbers
************
14 54 74 58 78 52 14 36 14 44

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit
Enter your choice: 3

Subscripts of odd numbers
*************************
0 2 4 5 10 12 13 16 17 18

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 4

The numbers with even subscripts
********************************
25 25 45 74 78 59 145 36 475 25

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 5

Minimum number
***************
14

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 6

Subscript of maximum number
***************************
16

DISPLAY
1. All numbers
2. Even numbers
3. Subscripts (indices) of odd numbers
4. The numbers with even subscripts
5. Minimum number
6. Subscript of maximum number
7. Exit

Enter your choice: 7
```

INSTRUCTIONS FOR UPLOADING YOUR ANSWERS:

1. **Make sure you have saved all your work** and exit from Microsoft Visual Studio 2017

2. Upon exit, if you hadn't saved already then Visual Studio will notify you to save it automatically; say **yes** to this.

3. Navigate into the directory in which you had created your lab guide solution and reverse click onto the **LG7_Sols** folder in there.

4. From the options menu, hover your mouse cursor over the **7-Zip** option and select "**Add to LG7_Sols.zip**" option to archive and compress your solutions folder. Change the name of the resulting archive to your name and surname to the zip file, i.e. **NameSurname.zip**

5. Upload the zip file to the instructor's PC by using your preferred browser;

   – **CTISL1: http://lab2t**

   – **CTISL2: http://lab7t** Inform your assistant that you have completed the upload process.

6. After your assistant's **approval**, delete your files using the "**Clean**" module you can either find in your start menu, the C: drive root folder or download through http://lab2t for Lab2, http://lab7t for Lab7.